

Tides and Their Effects

Chris Garrett and Leo R.M. Maas

The great Master of Philosophy drowned himself, because he could not apprehend the Cause of Tydes; but his Example cannot be so prevalent with all, as to put a Period to other Mens Inquiries into this subject.

—Richard Bolland, 1675



One may be surprised by this mythical account of Aristotle's reaction to the complicated currents of the Euripus in Greece, and assume that "other Mens Inquiries" have long since solved all the problems of tides in the world's oceans. But much of our under-

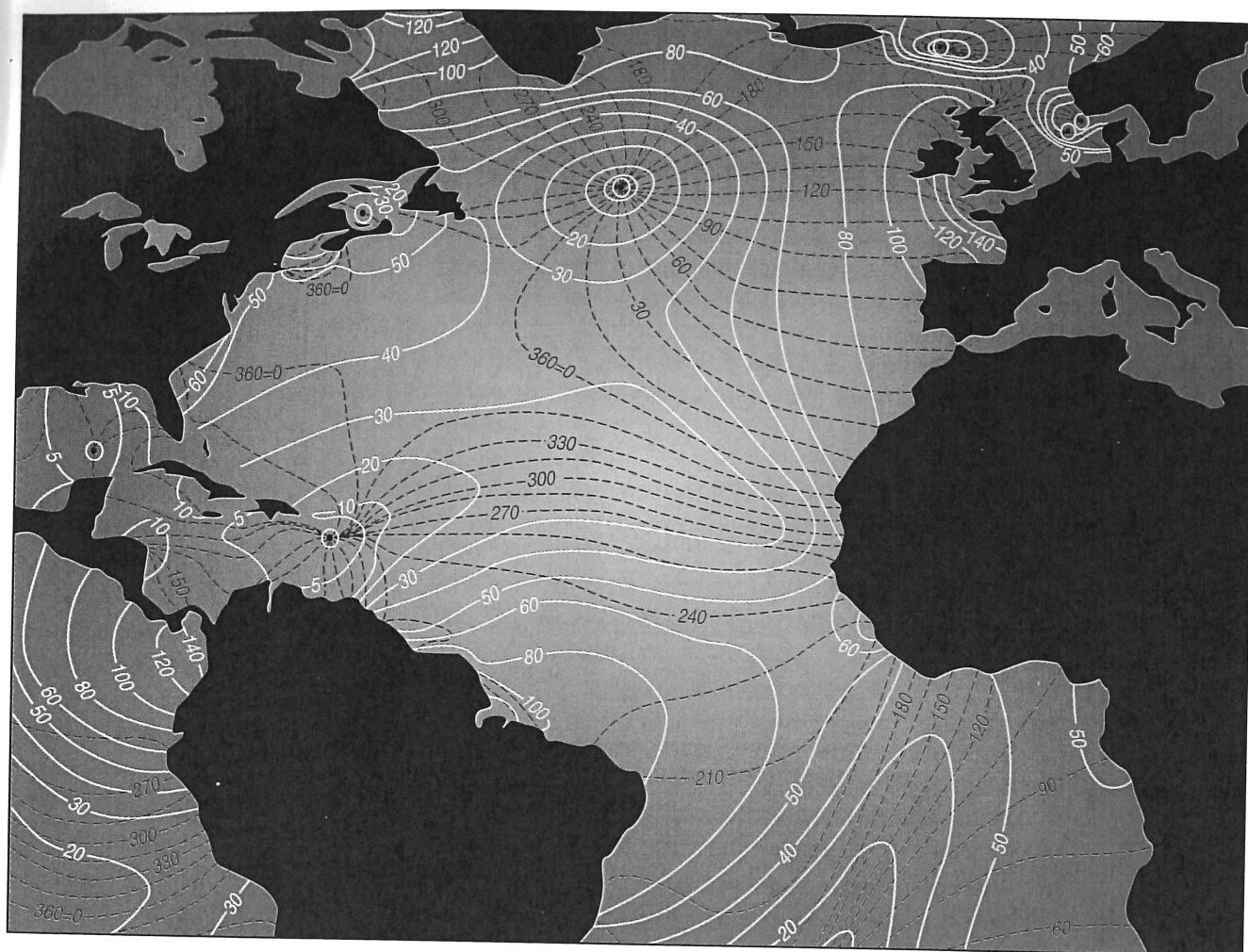
standing of the effects of tides in the coastal ocean is recent, and some important puzzles remain. Interest in tidal problems ebbs and flows like the tides themselves, but it is now fully recognized that tides do more than alter the water level in harbors; they affect many facets of circulation and mixing in continental shelf seas.

Tidal Theory is Based on Newton's Theory of Gravitation

Isaac Newton's 1687 theory of gravitation provides the basis for tidal theory: The earth and moon rotate around each other, with centrifugal force and gravitational attraction balancing one another. However, the gravitational force, which decreases with increasing distance, is greater than average on the side of the earth facing the moon, and less than average on the opposite side. If the earth were entirely covered by water, the ocean would bulge up both beneath the moon and at the antipodean point on the opposite side of the planet. As the earth rotates, any given location experiences both of these bulges, leading to two high tides a day. Unless the moon is over the equator, these tides are generally of different heights. As the moon makes its approximately 27.3-day revolution around the earth, it moves about 13° each day. The earth's daily rotation in the same direction requires an extra 50 minutes to match the moon's 13° movement, so high tides generally occur every 12 hours and 25 minutes.

The sun also exerts tide-producing forces at the earth's surface. Newton's law maintains that gravitational force is directly proportional to the mass of the bodies—the larger the body, the greater the force—and inversely proportional to the square of the distance between them—the

Tides affect many facets of circulation and mixing in continental shelf seas.



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The Atlantic 12.4-hour tide, as predicted by E.W. Schwiderski's numerical model, resembles waves rotating counterclockwise (with the dashed lines joining places where high tide occurs at the same time—the numbers are the phase lag, 30 degrees represents a delay of about 1 hour). The amplitude (solid lines, in centimeters) increases outwards from amphidromic points where there is no tide.

greater the distance, the smaller the force. This explains why the sun's gravitational pull on the earth is only about 46 percent that of the moon: though larger than the moon, the sun is much farther away. The main solar tide has a period, or repeat time, of 12 hours.

The actual tidal forcing is made up of these two main "constituents," plus many others that are associated with details of the orbits of the earth, moon, and sun. For example, there is an important constituent with a period of 12.7 hours associated with the elliptical orbit of the moon. This forcing, which is 19 percent of that of the 12.4-hour forcing, adds to it at perigee (when the moon is closest to the earth) and accounts for the big tidal forces then. Two weeks later, however, the two forces oppose each other to account for the weaker forcing at apogee (when the moon is farthest away from earth).

Continents greatly complicate the tides, but we can predict tidal rise and fall at any one place using past tidal records because the periods of the forcing constituents are not affected. Coastal tides have been measured since 1831 by the rise and fall of a float inside a "stilling well" connected to the outside water by an orifice small enough to suppress wind-generated waves. Once the constituents have been separated by mathematical analysis of the record, they can each be projected into the future and recombined to make a prediction of future water levels.

Resonance

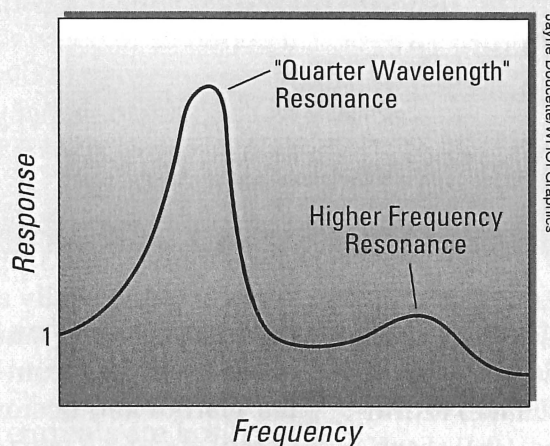
The concept of resonance can be illustrated by analogy with an AM radio. The signal from the broadcasting station has a particular frequency, and the internal circuitry of the radio has a natural frequency that can be adjusted by turning a knob. Beginning at the low end, as this internal frequency approaches the transmitter frequency, one begins to receive a signal, with maximum response (resonance) when the two frequencies match, and a diminishing response again as the radio is tuned to a higher frequency. Note that while the amplitude of the response varies, its frequency is always that of the transmitter, or forcing function. The radio is designed to have a very narrow resonance peak to avoid interference.

The ocean basins may also have various natural oscillations that are close to resonance with the direct tidal forcing, but with broad resonance peaks because of friction. The response depends on the difference between the natural and forcing frequencies, on friction, and also on how well the shapes of the forcing and the natural oscillations match.

For a bay excited by tidal forcing from the open ocean, the response is unity at very low frequency, for which the water has time to

reach the same level everywhere, and very small at high frequencies, at which the bay does not have time to respond. Resonance may occur at several frequencies in between; the most important, known as the "quarter-wavelength resonance," occurs when the tide advancing into the bay is reflected and returns to the entrance just in time to receive another push from the next oncoming tide.

The resonance peak is high and sharp if the bay is well separated from the ocean by a narrow entrance or major depth change and has little seafloor friction; the peak is lower and broader if the separation is less marked, or if there is significant friction.



High water for each constituent tends to circulate around "amphidromic points" where there is no tide, and bulge around ocean basins. Larger tides occur near the coasts. The tide may be further amplified on continental shelves by shallow-water effects that can include a near-resonant response in some regions. The world's highest tides, for example, occur near the head of the narrow, funnel-shaped Bay of Fundy, where the vertical range from low to high water can reach 16 meters. This is largely a consequence of a near-resonant response (see Box) of the Bay of Fundy and Gulf of Maine, which is a separate system from the North Atlantic because of the 20-fold change in depth from shelf to deep ocean. It has a natural period of about 13.3 hours, close enough to the 12.4-hour tidal period to respond vigorously to the Atlantic forcing.

The other important lunar constituent, with a period of 12.7 hours, is even closer to resonance and overtakes the 12-hour solar tide in importance. A consequence of this is that the tides in the region vary more over the one-month apogee-perigee cycle than over the two-week spring-neap cycle.

Spring-Neap Variations, the Age of the Tide, and Nodal Modulation

Variation in the tides from springs to neaps is well known; a conventional explanation is that large, spring, tides occur at full or new moon, when the moon and sun are aligned and pulling together, and small neap tides occur at the moon's quarters, when the moon and sun pull at right angles. However, a cursory examination of tide tables in most places with semidiurnal (twice-a-day) tides shows that the highest tides in a month tend to occur a day or two *after* the full or new moon! This lag, known as the "age" of the tide, was first recorded in 77 A.D. by Pliny the Elder, who attributed it to "the effect of what is going on in the heavens being felt after a short interval, as we observe with respect to lightning, thunder, and thunderbolts." Unfortunately this delightful theory has been discredited by recognition that light and gravity travel at the same speed. Modern theories relate the age of the tide to the near-resonant response of ocean basins themselves, but with the response limited by seafloor friction and the spatial mismatch of the response and the tidal forcing.

Tidal forcing and response are also modulated on longer time scales. One intriguing variation, known as the "nodal modulation," occurs over a period of 18.6 years and is associated with the moon's varying declination (movement of the moon's orbit toward the Equator). The total tidal forcing does not change, but its split into semidiurnal and diurnal (once a day) forcing does. This means that in most locations where the semidiurnal response dominates, the tide follows the modulation of the 12.4-hour forcing, and the next maximum will be in 1997. The effect is small (only about 3.7 percent), and tide predictions based on just a few years of data usually assume that the semidiurnal response will be modulated by the same amount. However, an analysis of some 50 years of sea-level data from Saint John on the Bay of Fundy, and 20 years from Bar Harbor and Boston on the Gulf of Maine, reveals an actual modulation of about 2.4 percent. This subtlety actually helped to support the theory and modeling that attributed the high tides of this region to a near-resonant response, but with allowance for bottom friction, which depends on the square of the current and so helps to reduce the effect of increased forcing.

Tidal Power is Expensive to Harness

The success of the theory in accounting for natural changes in tides increased confidence in predicting changes that would accompany Canadian development of tidal power in the Bay of Fundy's upper reaches. The simple expectation that barriers would effectively shorten the bay and bring it closer to resonance, with a consequent increase in the regional tides, was borne out in detailed modeling by David Greenberg (Bedford Institute of Oceanography). The predicted increase for the largest scheme discussed (with a proposed capacity of about 5,000 megawatts) was about 10 percent at Boston. Although a small change on its own, this is enough to slightly increase the possibility of flooding from tide and storm surge together. (Canadian scientists had some fun with proposals to tune the Bay of Fundy to resonance, generate tidal power, and then sell it south of the border to Bostonians wanting to

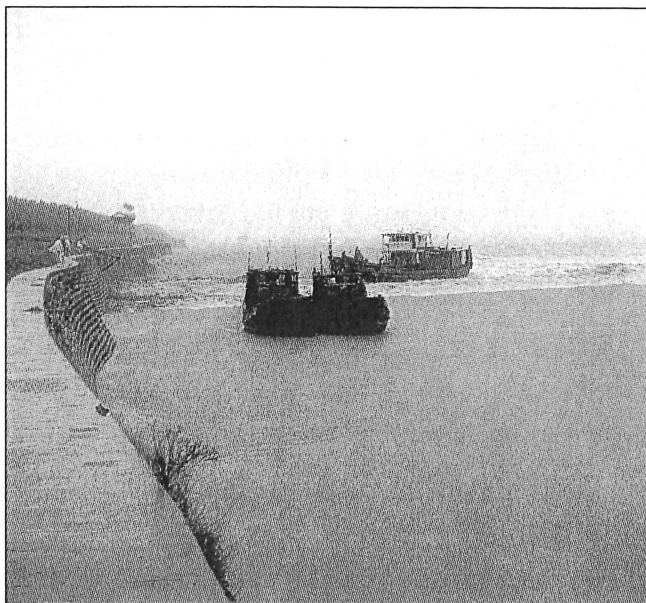
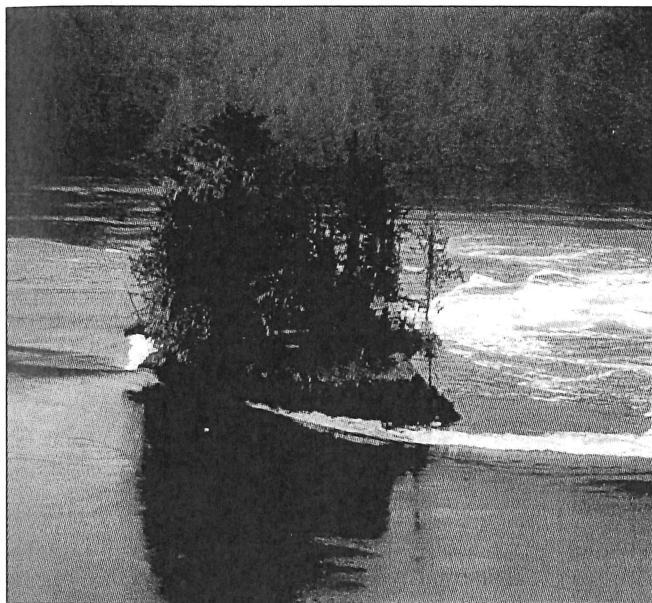
Canadian scientists had some fun with proposals to tune the Bay of Fundy to resonance.

Michael Woodward

pump out their flooded basements! They claimed that it represented a new export market and revenge for acid rain all in one.)

A 240-megawatt tidal power plant built nearly 30 years ago at La Rance in Brittany, France (see the photo on page 14), is still in operation, and a 20-megawatt pilot plant was built at Annapolis Royal in Nova Scotia in 1983, but plans for the big Fundy schemes, and others around the world, have receded for the present, mainly because construction costs for these facilities are so high that it would take many years to reap a return on the investment.

Michael Woodward



C. Garrett

Topography, Water Depth, and Other Factors Influence Tidal Currents

While tides are vertical changes in water level, tidal currents are horizontal water movements that are caused by the varying tides. At a particular location, tidal currents may be predicted from past records in the same way as tides themselves are. Relying on this to prepare current atlases is unsatisfactory, however, because tidal currents vary over short distances due to changes in topography. With increasing computer power, there has been a tendency in many parts of the world toward developing numerical models based on the laws of physics. These express little more than the tendency for water to flow downhill, while being pulled to the right (in the Northern Hemisphere) by the Coriolis force and slowed by bottom friction. Many models deal only with the depth-averaged current, but theories and observations both demonstrate that the current decreases toward the seafloor, with an interesting feature: If the current at each depth is split mathematically into one part that rotates clockwise in time and another that rotates counterclockwise, the latter increases more rapidly away from the bottom and tends to dominate near the seabed.

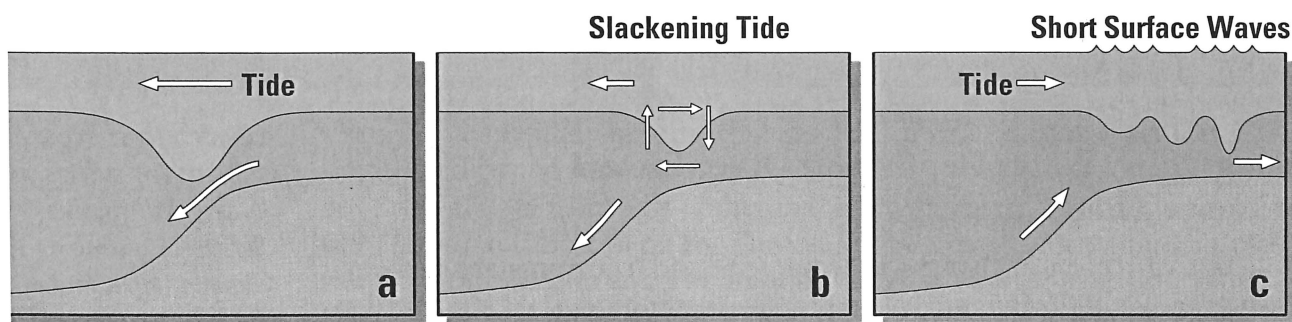
Tidal constituents (or frequencies) other than those present in the tidal forcing result from the nonlinear nature of friction that tidal currents experience at the seafloor, and from the advective nature of tidal currents. Tidal currents not only advect (transport) plankton, pollutants, and other floatables, but, if for some reason (usually related to variable

A 7-meter-per-second tidal current rushes in Nakwakto Rapids, British Columbia, Canada (left), and a tidal bore advances up the Qian Tang River in China (right).

Schematic of internal tide generation (a), its subsequent propagation onto the shelf (b), and its eventual breakup into a number of solitary waves (c). Thin arrows indicate water motion, thick arrows the progressing wave. Changes in surface wave pattern occur at locations that depend on factors such as the wind direction.

water depth) the tidal currents show a variation in strength or direction, this variation itself is advected by the tidal current. This is the prime nonlinear effect of fluid flow. The appearance of "higher harmonics" (whose frequencies are a multiple of that of the basic, say, semidiurnal tide, or whose periods are a fraction of that of the tide) is the first indication of the importance of this advective nonlinearity. Its effect is most pronounced in shallow water, where an advancing high tide tends to steepen, leading to the production in some rivers of a tidal bore, a wall of water much like surf approaching a beach. This same tendency is present, though at a much larger scale, in the interior of a stratified ocean when the depth-averaged current flows over abrupt topography.

If we assume the ocean consists of two layers (a shallow, warm, fresh layer atop a deep, cold, salty layer), a tidal current flowing off-slope produces a depression in the interface. This depression, like an inverted bulge of water, tries to flatten itself by propagating both on- and off-shelf. However, when the basic tidal current is strong enough, the bulge is unable to propagate against it and is arrested above the sloping region. Once the tide slackens, this shelfward-propagating wave is no longer checked, but moves onto the shelf and evolves into an "internal bore." The internal bore usually breaks down into a number of solitary waves in which the nonlinear tendency to steepen is exactly balanced by the tendency to disperse. This breakup of the tidal bore into successively smaller solitary waves is intimately related to the fact that the speed of



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these waves is proportional to their amplitude. The surface currents associated with these internal solitary waves modify any short wind waves present, and render them visible, even from space. Their eventual decay helps to provide energy for the upward mixing of nutrients that marine life depends upon. In this way the combination of tide and topographic slope exercises a curious form of remote control on biological production.

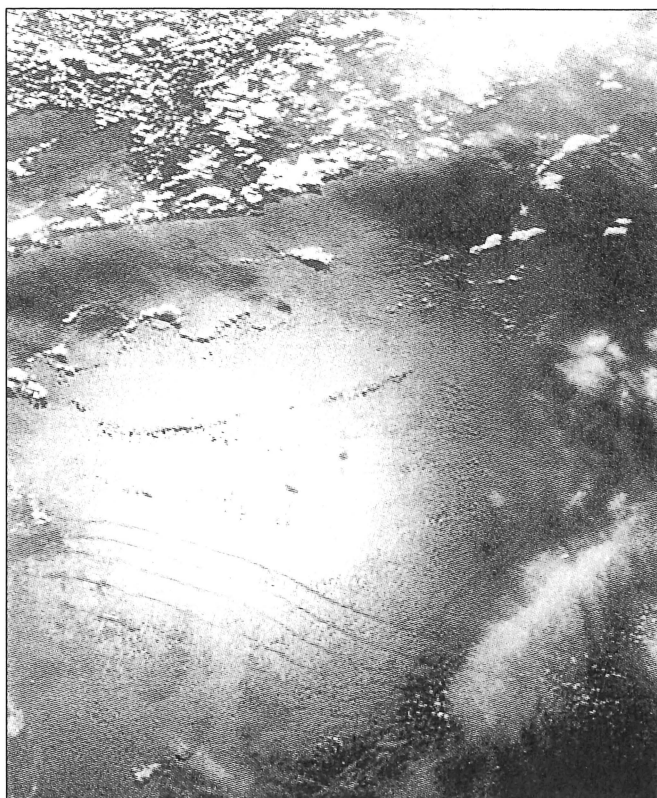
Mean Current Generation and the Dispersion of Matter

The advection of spatial variations in the tidal current by the tidal current itself not only generates higher harmonics (and solitary waves) but also produces a mean current. Unlike oscillatory tidal currents, a mean current can cause the net displacement of a fluid parcel over a tidal period. This is exemplified by a tidal flow's acceleration on passing a headland; the tidal flow enters the headland region smoothly, but continues as a jet, leaving an eddy in the lee of the headland. The opposite flow structure appears in the reverse tidal phase, so that residual

vortices will on average be found on both sides of the headland. An example of this, recently studied by Richard Signell (US Geological Survey) and Rocky Geyer (Woods Hole Oceanographic Institution) occurs at Gay Head on Martha's Vineyard. The mean flow in this case is usually localized, being restricted to the coastline's irregularities, hence there is no large-scale transport of the water and anything in it.

However, work initiated by John Huthnance (for his Ph.D. at Cambridge University) in the UK and applied to Georges Bank by Canadian oceanographer John Loder (for his Ph.D. at Dalhousie University) shows that extensive mean currents are produced by tidal flow over submarine banks and the steep slope of the continental shelf edge. The physical mechanisms and associated mathematical theories are quite complex (see the Box on page 33), but produce a mean current that in the Northern Hemisphere has the shallow water to the right facing downstream. This mean current at a fixed position, as would be measured by a moored current meter, is called the Eulerian mean current (for the 18th-century Swiss physicist Leonhard Euler). A fluid particle may experience an additional "Stokes drift" (named for the 19th-century English physicist Sir George Stokes) due to spatial variations in the magnitude of the oscillatory tidal current. For example, if the tidal motion is more or less circular, but increases in magnitude over shallower depths, a fluid parcel will be advected more strongly over the shallower regions and hence will acquire a net displacement. The total motion of a fluid particle is called the Lagrangian mean current (after the 18th-century French physicist Joseph Lagrange) and is related to the Eulerian mean current observed at a fixed position in space by a relationship that is popularly abbreviated as "Lagrange = Euler + Stokes." (This should not carry the implication that a Frenchman is a match for a Swiss citizen and an English gentleman combined!) On the sides of Georges Bank, the Stokes drift is in the opposite direction to the Eulerian mean current and only about one-third as big, so that the mean Lagrangian, or particle, motion is in the same direction as the Eulerian mean flow, clockwise around the Bank, but at about two-thirds the rate.

The mean current magnitude is dependent on the ratio of the width of the sloping region to the distance that a particle travels in half a tidal cycle (the tidal excursion, approximately 14 kilometers for a tide of 1 meter per second), and is strongest when these two length scales match. There appears to be a worldwide tendency for sandbanks generated in shallow seas to meet this requirement and also to be oriented with respect to the tidal current, such that the two residual current generating mechanisms (see the Box) are in tune. (In the Northern Hemisphere this requires a counterclockwise turning of these elongated underwater

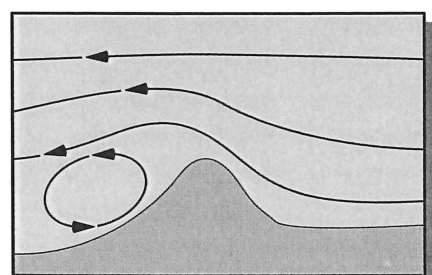
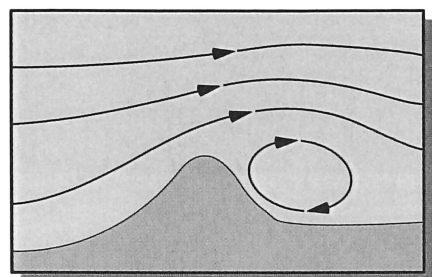
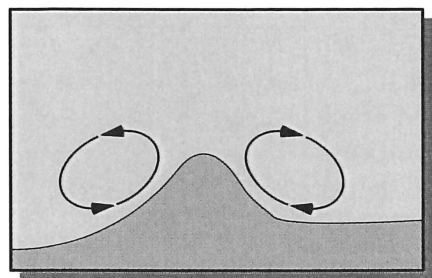


Surface traces of a series of internal solitary waves in the Andaman Sea (depth about 1,400 meters), west of Thailand, as seen from the Apollo-Soyuz spacecraft. North is to the left. The 100-kilometer-long solitary waves, which are visible at the sea surface due to the effect of the associated currents on short surface waves and hence on sun glitter patterns, propagate towards the west-coast of Thailand in the upper part of this photograph. They are about 15 kilometers apart and originate from the tides flowing over a sill located some 300 kilometers away. White regions are clouds.

dunes with respect to the dominant tidal current direction.) This suggests the existence of some interesting but not yet fully resolved feedback mechanism between a sandy bottom and the tidal flow above it, each "shaping" the other.

Although one might anticipate that the existence of a mean flow over a feature like the continental shelf slope could contribute greatly to the dispersion of matter, this would normally not be expected to occur in regions where the mean-flow streamlines are closed. However, work

largely pioneered by Jeff Zimmerman (Netherlands Institute for Sea Research) demonstrates that irregular particle paths may well result in this circumstance, even from regular velocity fields that lack any turbulent component. To appreciate this, consider a sequence of residual cells (generated by either of the processes discussed) in which particles, in the absence of a tidal current, would just circulate around their centers, following the streamline pattern. Then, when a periodic (tidal) current is superimposed on this, it might advect particles from one cell to the next. Depending on the relative strength of the tidal current to the residual current, particles may be advected back to the original cell during the reverse tidal phase, so that their mutual distances are not greatly affected. However, when some particles manage to stay in the neighboring cell (or even further away), their distance from some originally close-by particles is substantially increased, a process known as "chaotic stirring."



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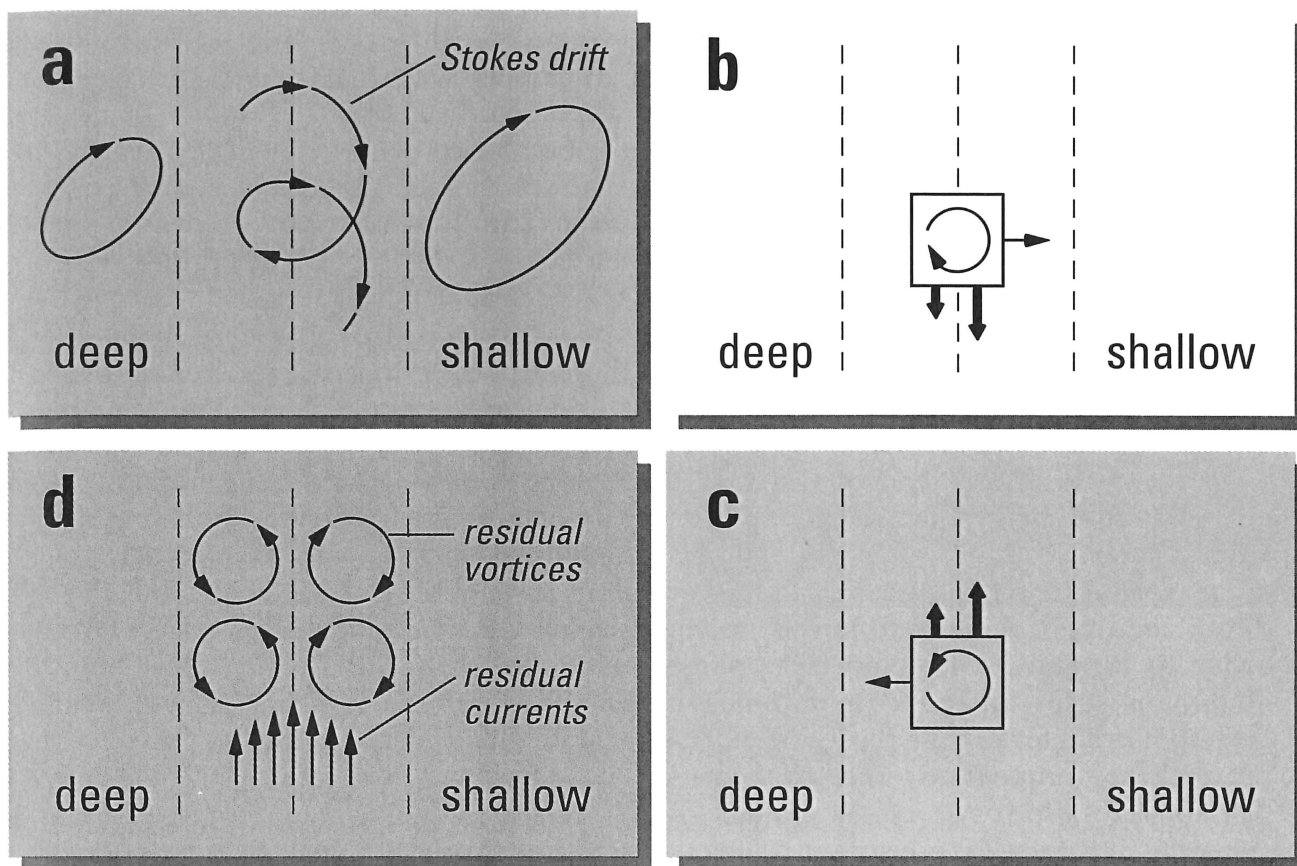
The net (mean) flow pattern of two phases of tidal flow past a headland. The water approaches the headland smoothly, but leaves it as a jet, creating a vortex in its lee.

Tidal Mixing

Turbulence generated at the seafloor by the tidal currents can exert important influences on the average density structure of the coastal ocean. In spring and summer this vertical mixing competes with the stratifying influence of surface heating. As recently as 1974, John Simpson and John Hunter (University College of North Wales) pointed out, on energetic grounds, that a very good indicator of the winner of this competition is

the single parameter (which now bears their names) of water depth divided by the cube of the maximum tidal current. For large values of this parameter, the water can become stratified like a lake, whereas for small values it is kept well-mixed throughout the summer. For the shallow seas around England, for the Fundy/Maine system, and for many other shelf areas, the dividing value is about 70 seconds cubed per square meter, so that a 1-meter-per-second tidal current can keep 70 meters of water well mixed. The effect shows up in oceanographic data and also in sea-surface temperature measurements made from space (on clear days) by observing the infrared emissions of the sea surface; well-mixed areas such as the Bay of Fundy and Georges Bank are kept cold at the surface by the mixing, though the shallowest parts of the mixed regions, such as the center of Georges Bank, may still heat up more than their surroundings as there is less water to heat.

Tidal mixing in shallow seas is clearly of biological importance, because it promotes productivity by returning nutrients to surface waters—unless the water is so deep that phytoplankton receive insufficient



Jayne Doucette/WHOI Graphics

light as they, too, are mixed up and down. In this respect, frontal regions between tidally mixed and stratified regions are of particular interest: Nutrients mixed to the surface on the side of the front where the nutrient level is high but the average light available to vertically mixed phytoplankton is too low for growth may be stirred across the front into the stratified region, where nutrients have been depleted but phytoplankton can remain in the well-lit upper part of the water column.

Global Importance

Seafloor turbulence associated with tidal currents dissipates a considerable amount of energy. For the Fundy/Maine system calculations based on models and data show this to be about 50,000 megawatts. The value is less well-established for other shallow seas, though it is known to be large in the seas around Britain and off the Patagonian coast of Argentina. However, the total global dissipation rate has become increasingly well known in recent years through a variety of estimates. The most direct of these calculates the rate at which tidal forces work against the vertical velocity of the sea surface; this is known from satellite altimetric mapping of the sea surface as well as from numerical models of open-ocean tides. According to estimates by David Cartwright and Richard Ray (NASA Goddard Space Flight Center), the global energy dissipation is 2.55 million megawatts for the 12.4-hour lunar tide, and about 20 percent higher for all the lunar tides. Corresponding to this, the moon is moving away from the earth by about 37 millimeters each year, a rate that has been confirmed to within 5 percent by James Williams, Skip Newhall, and Jean Dickie (Jet Propulsion Laboratory) using measurements

Tidal current ellipses and the resulting Stokes drift (a); residual current generation by tidal flow over a sloping bottom due to differential Coriolis forces (b, c) (two phases of the tidal flow); and (d) the net vorticity and mean flow pattern. Thin arrows indicate the water motion; thick arrows indicate the Coriolis forces acting on the sides of the fluid parcels. As the generation process is independent of the position along the slope, the residual current is parallel to it.

Residual Currents

Eulerian "residual currents" (the currents which result after the purely oscillatory part has been mathematically removed) can be generated in the vicinity of a sloping bottom by differential Coriolis and differential bottom frictional forces. Both of these forces vary depending on the tidal-current speed that a fluid particle experiences at its sides when these sides are over different water depths. As an example, consider a fluid particle that is advected upslope by the tidal current. Its front side, being over shallower depth, is moving slightly faster (the shallower depth requires the flow to speed up, in order to be able to transport the same amount of water) and hence experiences a larger Coriolis force than the rear side. The net clockwise torque experienced by the particle translates into a net clockwise rotation (or "vorticity") that itself is advected upslope in this phase of the tidal flow. The reverse process—the generation of counterclockwise vorticity and downslope transport of it—appears in the off-shelf phase of the tidal current. Averaged over a tidal cycle, then, a pair of counter-rotating steady currents are generated. This happens at every point along a uniform along-shelf bottom profile, and leaves the residual, or mean Eulerian, current flowing parallel to

the bottom contours with (in the Northern Hemisphere) the shallow side to the right when facing downstream. For a current rotating clockwise (as is usual for Northern Hemisphere currents) the Stokes drift will be in the opposite direction to this. Hence, in this case, the total particle drift velocity, the Lagrangian current, will be weaker than the local Eulerian current. The opposite will occur for a current rotating counterclockwise. The actual production of the Eulerian mean flow is subtly dependent on the presence of bottom friction, since in its absence the generation of, for example, clockwise vorticity and its subsequent advection, are exactly out of phase, and therefore would vanish when averaged over a tidal period. The presence of friction provides the necessary phase shift between the two above processes, leading to a nonzero residual current. Remarkably, the generated residual current is strongest when the frictional coefficient is weakest (but remains nonzero)! Apart from this catalyzing role, (differential) bottom friction acting on a tidal flow may also directly drive a mean flow all by itself; this may add to or subtract from the previously discussed mean flow, depending on the orientation of the tidal current with respect to the depth contours.

of the changes in travel time of a laser beam bounced off reflectors first placed on the moon by *Apollo* astronauts in 1969.

The solar tides add a further 20 percent or so to the tidal dissipation, which totals about 3.6 million megawatts altogether. This decreases the earth's rotational energy, giving a 1-second increase in the length of the day every 41,000 years. A major remaining oceanographic puzzle concerns whether the dissipation occurs mostly in shallow seas, or whether a significant fraction involves deep-ocean generation of tidal-frequency internal waves that travel into the stratified ocean interior and cause turbulent mixing.

While the phenomenon of tides has been known as long as the shores have been inhabited, its multifaceted effects only slowly unfold. This makes for a fascinating research topic, with ever further-reaching environmental applications. ☉

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Chris Garrett spent his first undergraduate year in a room that had been Isaac Newton's attic, but chiefly remembers the lack of heating in England's coldest winter for 80 years. He stayed at Cambridge for his Ph.D., but then defected to the west, spending a year at the University of British Columbia and two years at Scripps Institution of Oceanography (University of California, San Diego) before moving to the Department of Oceanography at Dalhousie University in Halifax, Nova Scotia. His scientific interests have largely moved on from tides to questions of ocean mixing and air-sea interaction, and he occasionally enjoys working on more immediately practical topics involving unrelated things like icebergs and radioactive waste. After 20 years in Nova Scotia he recently returned to the Pacific southwest (Canada's), where he is at the School of Earth and Ocean Sciences and the Department of Physics and Astronomy at the University of Victoria, but he still welcomes invitations to visit Woods Hole....



Bedford Institute of Oceanography, Department of Fisheries and Oceans, Canada

Being born in an area of the Netherlands that has been frequently flooded in the past may have subconsciously raised Leo R.M. Maas's interest in the untamed behavior of the ocean. Consciously, an interest in natural phenomena that occur on time and length scales "that we live in" led him to study physical oceanography at the University of Utrecht. After his Ph.D. he continued his research on tides at the Netherlands Institute for Sea Research, albeit, now being behind safe dikes, taking an occasional detour to study some more elementary fluid dynamical and mathematical topics. Having never visited Woods Hole, his desire to receive an invitation to do so is, possibly, even greater than that of his co-author....

A false color image of sea-surface temperature (blue is cold, red is warm) in the Bay of Fundy and Gulf of Maine on 27 August 1984. Tidal mixing keeps the surface cold in the Bay of Fundy, over Nantucket Shoals south of Cape Cod, and over Georges Bank (though the center of this, like the upper reaches of the Bay of Fundy, is shallow enough to have become warm by late summer even though well-mixed). The rest of the Gulf of Maine is stratified and warm at the surface; the red ribbon to the south is the Gulf Stream, with a few patches of

A Tribute to Henry Stommel



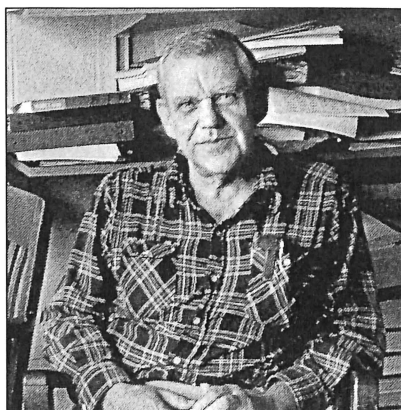
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